

Flavonoid permeability across an in situ model of the blood-brain barrier.

Understanding mechanisms associated with flavonoid neuroprotection is complicated by the lack of information on their ability to enter the CNS. This study examined naringenin and quercetin permeability across the blood-brain barrier (BBB), using in vitro (ECV304/C6 coculture) and in situ (rat) models. We report measurable permeabilities (P_{app}) for both flavonoids across the in vitro BBB model, consistent with their lipophilicity. Both flavonoids showed measurable in situ BBB permeability. The rates of uptake (K_{in}) into the right cerebral hemisphere were 0.145 and 0.019 ml min⁻¹ g⁻¹ for naringenin and quercetin, respectively. Quercetin K_{in} was comparable to that of colchicine (0.006 ml min⁻¹ g⁻¹), a substrate for P-glycoprotein (P-gp). Preadministration of the P-gp inhibitor PSC833 or GF120918 (10 mg/kg body wt) significantly increased colchicine K_{in} , but only GF120918 (able to inhibit breast cancer resistance protein, BCRP) affected K_{in} for quercetin. Naringenin K_{in} was not affected. The influence of efflux transporters on flavonoid permeability at the BBB was further studied using MDCK-MDR1 and immortalized rat brain endothelial cells (RBE4). Colchicine, quercetin, and naringenin all showed measurable accumulation (distribution volume, V_d (microl/mg protein)) in both cell types. The V_d for colchicine increased significantly in both cell lines following incubation with either PSC833 (25 micromM) or GF120918 (25 micromM). Both inhibitors also caused an increase in naringenin V_d ; by contrast only GF120918 incubation significantly increased quercetin V_d . In conclusion, the results demonstrate that flavonoids are able to traverse the BBB in vivo. However, the permeability of certain flavonoids in vivo is influenced by their lipophilicity and interactions with efflux transporters.